Laboratory Course: [BCS-351] Data Structures Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: CODE TEMPLATE**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Department of Computer Sc**

Starter Code Templates & Clean C Implementation Patterns

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1. Circular Queue Template Pattern (C Language)

- Structure: struct CircularQueue { int items[SIZE]; int front; int rear; };
- Enqueue logic: if ((rear + 1) % SIZE == front) -> OVERFLOW; else rear = (rear + 1) % SIZE;
- Dequeue logic: if (front == -1) -> UNDERFLOW; element = items[front]; if (front == rear) reset;

2. Binary Search Tree (BST) Node Template

- Node declaration: struct Node { int data; struct Node *left, *right; };
- Memory allocation: struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
- Recursive traversal: void inorder(struct Node* root) { if(root) { inorder(root->left); printf("%d ", root->data); inorder(root->right); } }